

Vector Space Model - Tutorial

Introduction to Information Retrieval

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Problem 1

- You are designing a search system where the collection to be searched has three fields: **TITLE**, **ABSTRACT** and **BODY**. For a given query, you want the search algorithm to weight the presence of the query terms in the **TITLE** field more than the presence in the **ABSTRACT** or **BODY**.
 - ① How to do this?
 - ② Is it necessary for all zones to use the same match function?

Problem 2

- Suppose in the previous problem, we adapt a **Boolean score function** for every field:
 - ▶ for a given query $Q = \{q\}$, the Boolean score function for a field in a document takes on the value 1 if the query term **q** is present in the field, and 0 otherwise;
 - ▶ we set the field weights as:
 - ★ $weight(T) = 0.45$
 - ★ $weight(A) = 0.35$
 - ★ $weight(B) = 0.2$
 - ▶ this corresponds to an application in which a match in the **TITLE** field is considered the most important while a match in **BODY** is the least.
- ? What are all the distinct score values a document may get?

Problem 3

- You are designing a search system where the collection to be searched has three fields: **TITLE**, **ABSTRACT** and **BODY**. For a given query, you want the search algorithm to weight the presence of the query terms in the **TITLE** field more than the presence in the **ABSTRACT** or **BODY**.
- ? What if you want to allow phrase search in the **BODY** field?

Problem 4

? Why IDF of a term in a collection is always finite?

Problem 5

Term	IDF	Doc1	Doc2	Doc3
car	1.65	27	4	24
auto	2.08	3	33	0
insurance	1.62	0	33	29
best	1.5	14	0	17

- Compute the tf-idf weights of all the terms, for each document.

Problem 6

$$IDF(t) = \log \frac{N}{|df(t)|}$$

- How does the base of the logarithm affect the relative scores of two documents for a given query?

Problem 7

- If we were to stem/case-fold **institution**, **Institution** and **Institute** to a common stem **institut** before setting up the vector space, how the values of tf and idf will be changed?

Problem 8

Term	IDF	Doc1		Doc2		Doc3	
		TF	TF-IDF	TF	TF-IDF	TF	TF-IDF
car	1.65	27	44.55	4	6.6	24	39.6
auto	2.08	3	6.24	33	68.64	0	0
insurance	1.62	0	0	33	53.46	29	46.98
best	1.5	14	21	0	0	17	25.5

- Compute the Euclidean normalized document vectors for each of the documents, where each vector has four components, one for each of the four terms.

Problem 8

Term	IDF	Doc1		Doc2		Doc3	
		TF	TF-IDF	TF	TF-IDF	TF	TF-IDF
car	1.65	27	44.55	4	6.6	24	39.6
auto	2.08	3	6.24	33	68.64	0	0
insurance	1.62	0	0	33	53.46	29	46.98
best	1.5	14	21	0	0	17	25.5

- Compute the Euclidean normalized document vectors for each of the documents, where each vector has four components, one for each of the four terms.

Term	IDF	Doc1		Doc2		Doc3	
		TF	Eu. Norm. W.	TF	Eu. Norm. W.	TF	Eu. Norm. W.
car	1.65	27	0.897	4	0.076	24	0.595
auto	2.08	3	0.125	33	0.786	0	0
insurance	1.62	0	0	33	0.613	29	0.706
best	1.5	14	0.43	0	0	17	0.383

Problem 9

- Assume there are 10 million documents in a collection (i.e. $N = 10,000,000$). Given the query $Q = \text{'digital camera'}$, compute the vector space similarity between the query Q and the document d having content: $\text{'digital camera and video camera'}$. Consider the weighting scheme: *Inc./ln*, that is, based on the SMART notation $d_{tf} d_{df} d_{norm} \cdot q_{tf} q_{df} q_{norm}$:
 - d_{tf} and q_{tf} - logarithmic term frequency weighting for document and query;
 - d_{df} and q_{df} - idf weighting for query only and not document;
 - Normalization** - cosine normalization for the document only.

Treat **and** as a stop word. What is the final similarity score of **d** for query Q ?

		document d				query Q				score
word	df_{word}	tf	d_{tf}	d_{df}	w_d	tf	q_{tf}	q_{df}	w_q	$w_d \cdot w_q$
camera	50,000									
digital	10,000									
video	100,000									

Problem 10

- In the data provide for the previous problem, compute the score of the same document if we use the weighting scheme: *Inc./tc*. What difference will be seen in the overall ranking of all the documents in the collection?